The Islamic University of Gaza

Facutly of Engineering

Software Programmig Lab

ECOM4024

Proposal Software Engineering Project

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Introduction

In an era of increasing academic demands, efficient time management has become a cornerstone of student success. **Study Planner** is a web-based solution designed to empower learners by transforming complex academic schedules into manageable, structured tasks. This project aims to leverage modern web technologies to provide an intuitive interface for tracking progress and optimizing study habits. The following proposal outlines the core functionalities, technical specifications, and the strategic roadmap for developing a comprehensive tool that streamlines the learning journey

Includes

1.Chosen Project Idea

The proposed project is a Smart Study Planner Web Application designed to help students effectively .organize their academic tasks, manage their time, and improve their study habits

Unlike traditional planners, this system provides an intelligent and automated approach to scheduling. Instead of requiring students to manually create their study plans, the platform asks users to input key information such as available daily study hours and task difficulty levels. Based on this input, the .system automatically generates a personalized and realistic study schedule

The application also includes a centralized dashboard that displays upcoming deadlines with countdown timers, helping students stay aware of their priorities. In addition, it offers smart .notifications through email or browser alerts to remind users of important deadlines in advance

To further enhance productivity, the system provides progress tracking features using visual charts that show completion rates for each subject. It also supports breaking down large assignments into .smaller, manageable tasks, making studying less overwhelming

Overall, the idea focuses on transforming the way students plan their studies by combining automation, organization, and user-friendly design into one integrated platform.

2.Justification: why this idea?

Because it solves a common problem for university students: managing time, assessments, and deadlines effectively

Unlike generic tools, this planner is tailored to academic needs, offering features such as automatic task distribution, GPA tracking, focus mode, and progress visualization. The idea is practical, impactful, and provides immediate benefits to students while giving the team an opportunity to apply core Software Engineering principles on a real-world problem

3. Problem Statement

Students often face stress and distraction due to the heavy load of courses, projects, and exams. Relying on paper schedules or simple apps forces them to plan manually, which results in unrealistic timetables, forgotten deadlines, difficulty breaking down large projects, and weak focus caused by constant notifications

The proposed study planner addresses these challenges by

Smart time management: Automatically distributing tasks based on available study hours and task difficulty

Deadline alerts: Sending reminders 24h, 6h, and 1h before submission

Progress tracking: Visual charts showing achievement levels per subject

Task breakdown: Dividing large projects into manageable parts integrated into daily plans

Focus mode: Pomodoro timer with automatic breaks and notification blocking

Outcome

The system transforms the student's experience from chaos and pressure to clarity and organization, providing realistic daily plans, early warnings, visual progress insights, and a distraction-free study environment

4.Target Users & Use Cases

Target Users*

The Study Planner system is designed for the following user groups:

University students who manage multiple courses, assignments, and deadlines .

School students who need help organizing their study schedules and exams .

Self-learners and individuals enrolled in online courses who require structured planning and time . management

The system focuses on users who face challenges in organizing their academic tasks and managing their time effectively

Use Cases*

The Study Planner system provides several key functionalities that allow users to efficiently manage their academic activities:

Task Management: Users can add, update, and delete study tasks and assignments, including details . such as deadlines and subject names

Study Schedule Creation: Users can create personalized daily, weekly, or monthly study plans based . on their academic needs

Task Tracking: Users can view pending and completed tasks to monitor their progress .

Notifications & Reminders: The system sends reminders for upcoming deadlines and important . academic events

Progress Monitoring: Users can track their overall study progress and completed tasks over time .

5. Brief Description of the Core Features

Our Study Planner web application focuses on providing a streamlined and efficient user experience through the following core features:

Customizable Academic Dashboard: A central interface that displays a summary of the student's daily schedule and upcoming deadlines at a glance.

Course & Subject Organizer: Enables students to add and manage their current courses, including instructor details and credit hours

Smart Task Management (To-Do List): Allows users to create, categorize, and prioritize academic tasks such as assignments, quizzes, and projects.

Progress Tracking & Visualization: Features visual progress bars and charts to show the completion percentage for each subject and overall semester goals

Automated Deadline Alerts: Integrated notification system to remind students of approaching task deadlines and exam dates

- **Study Session Timer (Pomodoro):** An embedded tool to help students manage study intervals and improve focus during preparation

Feature	Description	User Benefit
Dashboard	Visual summary of current academic status.	Quick access to daily priorities.
Course Manager	Centralized database for all enrolled subjects.	Better organization of academic content.
Task Tracker	Prioritized list of assignments and quizzes.	Prevents missing important deadlines.
Progress Bars	Visual representation of task completion.	Boosts motivation and time management.
Notifications	Automated web alerts for upcoming tasks.	Ensures timely submission of assignments.

6.Comparison with Existing Solutions

While there are many productivity and scheduling tools available, most of them are designed for general business or personal use, leaving a gap in meeting the specific needs of students. Below is a comparison between the proposed Study Planner and existing solutions:

Feature	Traditional Tools / Google Calendar) (Todoist	Complex Management Systems (Notion)	Our Proposed Study Planner
Input Method	**Manual**: User must set every time slot and date manually	**Customizable but Tedious**: Require high effort to build a system	**Automated:** Use enters tasks and deadlines; the system suggests the plan
Academic Focus	**Low**: Treats a Lunch Date" and a" Final Exam" with" the same priority	**Medium**: Can be customized, but lacks built-in academic logic	**High:** Specifically designed to handle course loads, ,credits and GPA
Rescheduling	**Rigid:** If you miss a slot, you must move it manually	**Manual:** Require updatingmultiple pages or databases	**Dynamic:** Automatically redistributes missed tasks into next available slots
Learning Curve	**Low:** Easy to use but limited in features	**High:** Takes weeks to master and "setup" a workspace	**Very Low:** -Streamlined, "one click" interface focused only on studying

Reminders	**Standard:** Simple alerts at a specific time	**Basic:** Notifications within the app	**Smart Alerts:** Intelligent reminders based on the urgency of the deadline

7.Expected Impact or Benefit

The proposed Study Planner project is expected to provide useful benefits for students by helping them organize their academic tasks and manage their time more effectively. Many students struggle with balancing assignments, exams, and personal responsibilities, which often leads to stress and missed deadlines. This system aims to solve these issues by offering a clear and structured way to plan .daily and weekly activities

Additionally, the Study Planner will enhance productivity by allowing users to prioritize their tasks and track their progress. Features such as reminders and scheduling will help students stay on track and .reduce the chances of forgetting important deadlines

Also, the project can improve students' overall academic performance by encouraging better study habits and time management skills. It will also provide a simple and user-friendly experience, making .it accessible for all students regardless of their technical background

In general, this project aims to make studying more organized, less stressful, and more efficient, leading to better outcomes for students. This project can also be expanded in the future to include smart recommendations and AI-based scheduling

8. Initial thoughts on the technology/tools that may be used.

The proposed Smart Study Planner will be developed as a web-based application using modern and practical technologies to ensure usability, performance, and scalability

For the front-end, HTML, CSS, and JavaScript will be used to build a clean and user-friendly interface. A framework such as React can be utilized to enhance interactivity and provide a smoother user experience

On the back-end, Node.js with Express will be used to handle the system logic, including task management, scheduling operations, and user authentication. This will allow efficient processing of user data and application workflows

A database such as MongoDB will be used to store user information, tasks, deadlines, and progress data in a structured and secure manner

The scheduling functionality will be implemented using a basic algorithm that distributes tasks based on user input such as available study time and task priority. Notifications and reminders can be implemented using tools such as Firebase or email services to ensure users are aware of upcoming deadlines

In future development, advanced technologies such as Artificial Intelligence can be integrated to improve scheduling accuracy and provide smarter study recommendations

Overall, the selected technologies are suitable for the project scope, easy to implement, and allow future expansion of the system